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Practical No. 01

Aim: Write a program to implement Linear Search and Binary search.

Software Requirement: Python IDE, PyCharm, Visual Studio Code (VS Code), Jupyter Notebook.

Theory:

Searching: Searching is just finding out the data item among given list. Two cases are possible that either item is found or item not present in the list. The Searching is divided into two types.

1) Linear Search

2) Binary Search

Linear search:

Linear Search is a very simple search algorithm. In this sequential search is done by searching items one by one. Each item is checked and if a match is found then that particular item is returned, otherwise the search is continuing till end of the list of items. If item which is to be searched is not present in list, then the search is called as unsuccessful search. Linear search is also known as sequential search.

Algorithm for Linear Search Algorithm:

The algorithm for linear search can be broken down into the following steps:

- **Start:** Begin at the first element of the collection of elements.
- **Compare:** Compare the current element with the desired element.
- **Found:** If the current element is equal to the desired element, return true or index to the current element.
- **Move:** Otherwise, move to the next element in the collection.
- **Repeat:** Repeat steps 2-4 until we have reached the end of collection.
- **Not found:** If the end of the collection is reached without finding the desired element, return that the desired element is not in the array.

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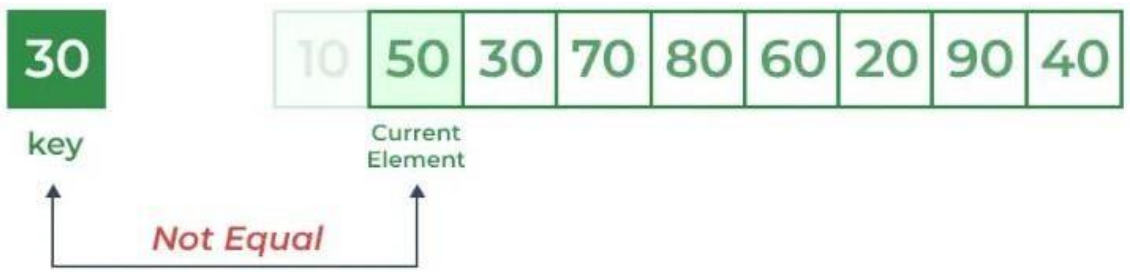
For example: Consider the array **arr[] = {10, 50, 30, 70, 80, 20, 90, 40}** and **key** = 30

Step 1: Start from the first element (index 0) and compare **key** with each element (arr[i]).

- Comparing key with first element arr[0]. Since not equal, the iterator moves to the next element as a potential match.

- Comparing key with next element arr[1]. Since not equal, the iterator moves to the next element as a potential match.



Step 2: Now when comparing arr[2] with key, the value matches. So the Linear Search Algorithm will yield a successful message and return the index of the element when key is found (here 2).

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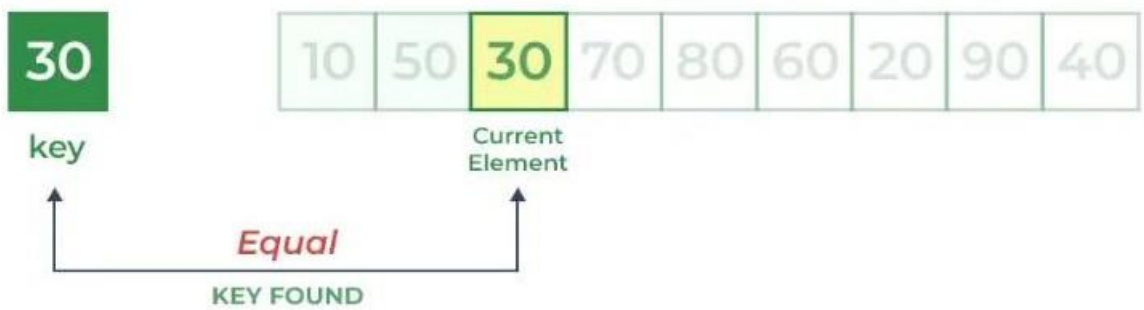

Advantages of Linear Search Algorithm:

- Linear search can be used irrespective of whether the array is sorted or not. It can be used on arrays of any data type.
- Does not require any additional memory.
- It is a well-suited algorithm for small datasets.

Disadvantages of Linear Search Algorithm:

- Linear search has a time complexity of O(N), which in turn makes it slow for large datasets.
- Not suitable for large arrays.

Binary Search:

Binary search is a search algorithm used to find the position of a target value within a **sorted** array. It works by repeatedly dividing the search interval in half until the target value is found or the interval is empty. The search interval is halved by comparing the target element with the middle value of the search space.

Binary Search Algorithm:

Below is the step-by-step algorithm for Binary Search:

- Divide the search space into two halves by finding the middle index "mid".
- Compare the middle element of the search space with the **key**.

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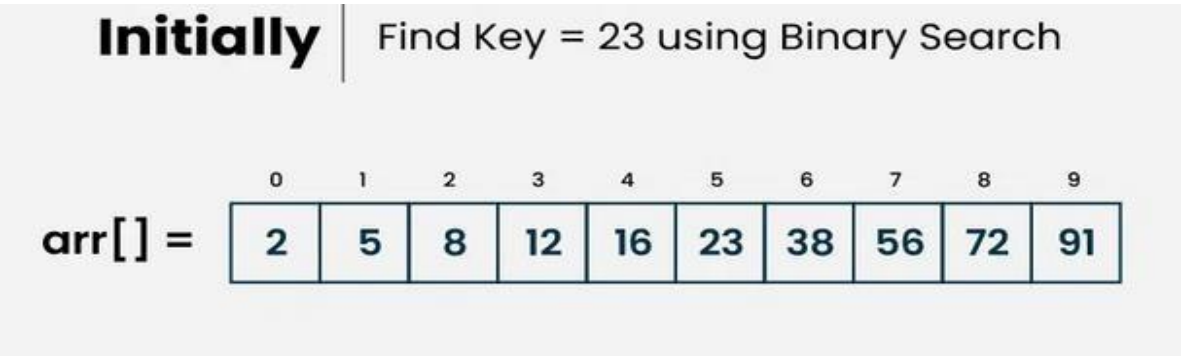
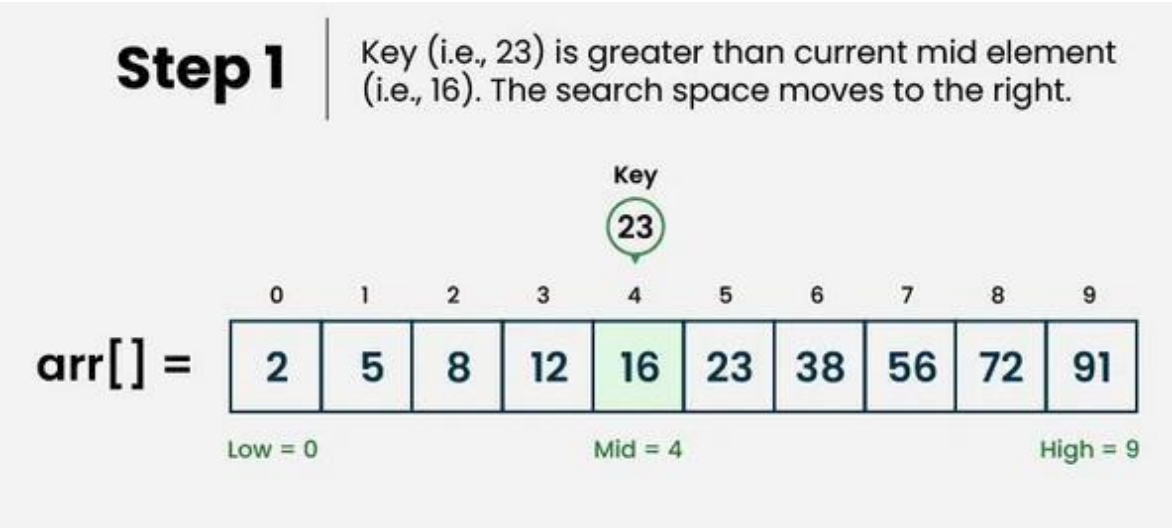
- If the **key** is found at middle element, the process is terminated.
- If the **key** is not found at middle element, choose which half will be used as the next search space.
  - If the **key** is smaller than the middle element, then the **left** side is used for next search.
  - If the **key** is larger than the middle element, then the **right** side is used for next search.
- This process is continued until the **key** is found or the total search space is exhausted.

Example:

Consider an array **arr[] = {2, 5, 8, 12, 16, 23, 38, 56, 72, 91}**, and the **target = 23**.





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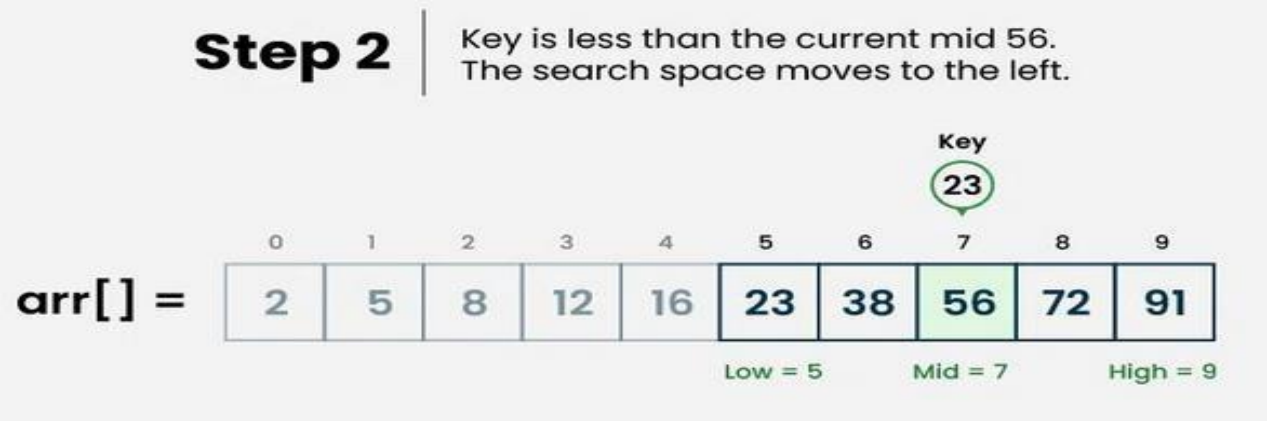

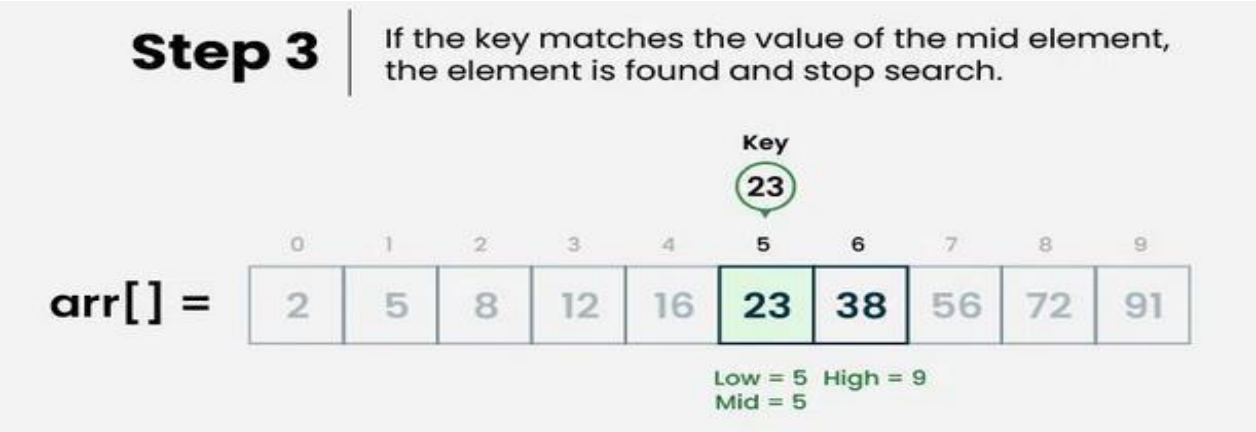

Advantages of Binary Search:

- Binary search is faster than linear search, especially for large arrays.
- More efficient than other searching algorithms with a similar time complexity, such as interpolation search or exponential search.
- Binary search is well-suited for searching large datasets that are stored in external memory, such as on a hard drive or in the cloud.

Disadvantages of Binary Search:

- The array should be sorted.

- Binary search requires that the data structure being searched be stored in contiguous memory locations.
- Binary search requires that the elements of the array be comparable, meaning that they must be able to be ordered.

Program:

For Linear Search:

```
# List of elements
arr = [1, 3, 5, 4, 7, 9]
key = 7
n = len(arr)
def linear_search(arr, n, key):
    # Searching list1 sequentially
    for i in range(0, n):
        if arr[i] == key:
            return i
    return -1
# Perform linear search
res = linear_search(arr, n, key)

# Check and print the result
if res == -1:
    print("Element not found")
else:
    print("Element found at index:", res)
```

OUTPUT:

Element found at index: 4

For Binary Search:

```
# Initial list
list = [12, 24, 32, 39, 45, 50, 54]
n = 45
def binary_search(list, n):
    low = 0
    high = len(list) - 1

    while low <= high:
```

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```
        mid = (high + low) // 2

        # Check if n is present at mid
        if list[mid] < n:
            low = mid + 1
        # If n is greater, search the right half
        elif list[mid] > n:
            high = mid - 1
        # If n is equal to the middle element, return its index
        else:
            return mid

    # Element was not present in the list
    return -1

# Function call
result = binary_search(list, n)
if result != -1:
    print("Element is present at index", str(result))
else:
    print("Element is not present in list
```

OUTPUT:

Element is present at index 4

Conclusion:

Hence, thus I have executed the Python program to perform Linear Search and Binary Search Successfully.